

Platform Software · Optical Tracking

SKADI Tracking Software (API and Viewer)

Start with evidence and follow the SKADI software layer from marker definition to application: Define → Open → Track → Apply.

Period 2023.02 - present | Evidence state Ongoing | Portfolio tier Platform Software

Maintain MarkerEditor, SKADI Viewer, and DtSkadi.dll as one integration flow, centered on medical surgical navigation and extended to robot precision docking.

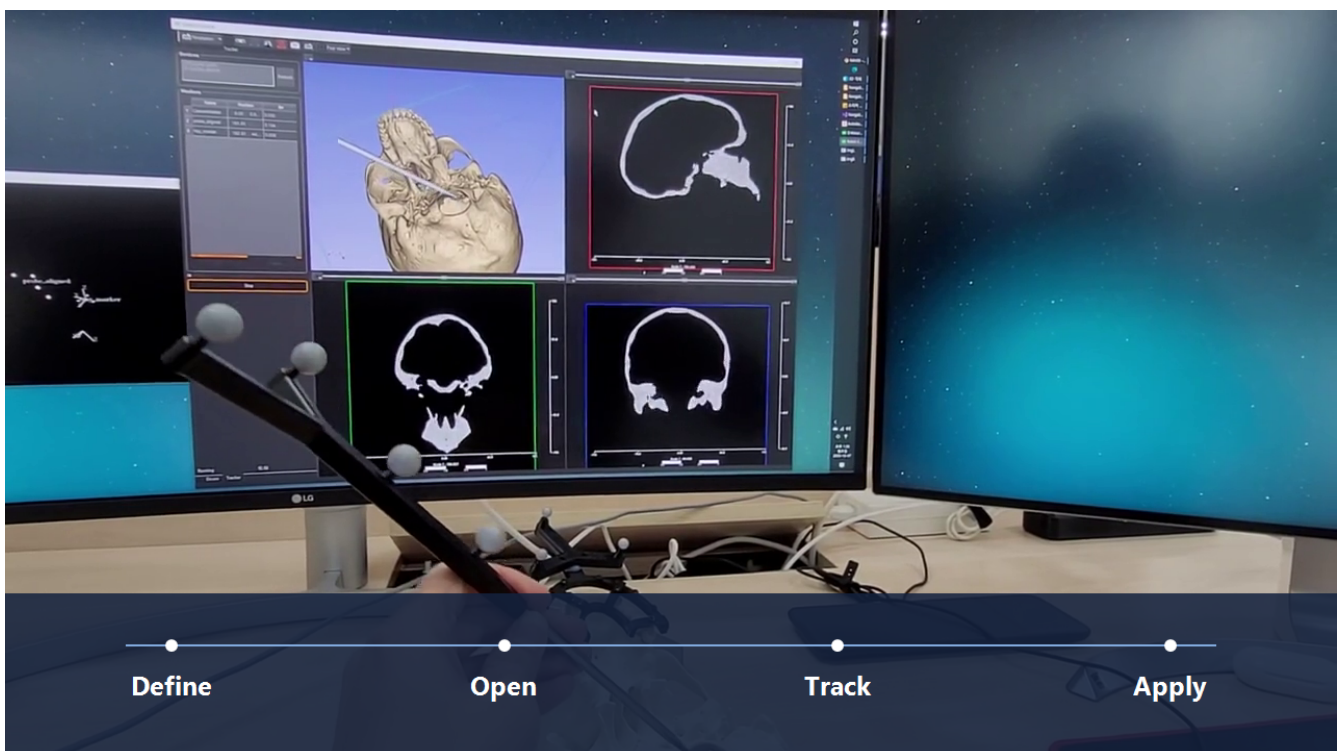


Figure 1. Silent lead video following Define → Open → Track → Apply through MarkerEditor, the API, Viewer 6DoF, and the medical and robot application boundaries.

Problem and application boundary

An optical tracker is not used as bare hardware. Developers need predictable interfaces for marker definition, API-open failures, real-time 6DoF tracking, and delivery of application transforms.

Marker definition and input validation

MarkerEditor imports, saves, and exports marker geometry and validates coordinate input.

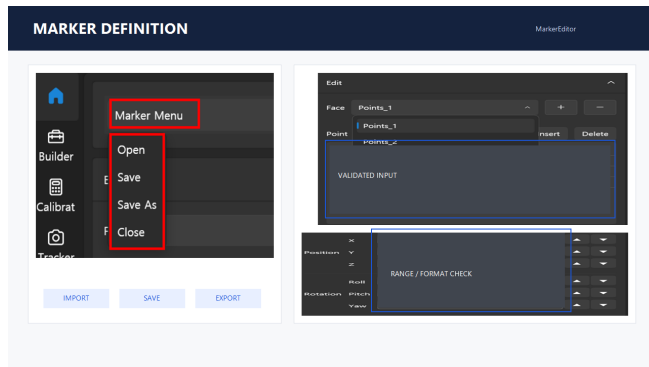


Figure 2. MarkerEditor marker-file import, save, and export flow with input validation

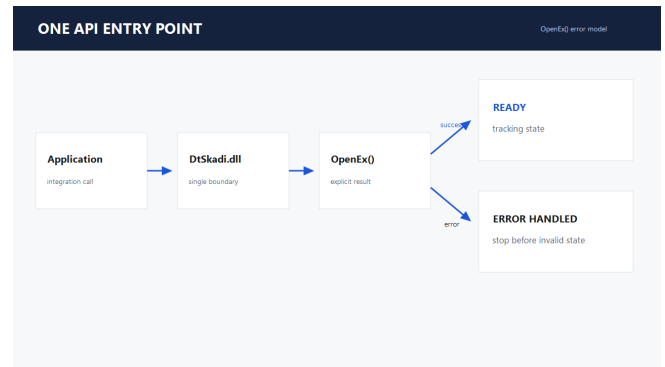


Figure 3. Single API entry point with OpenEx() success and error branches

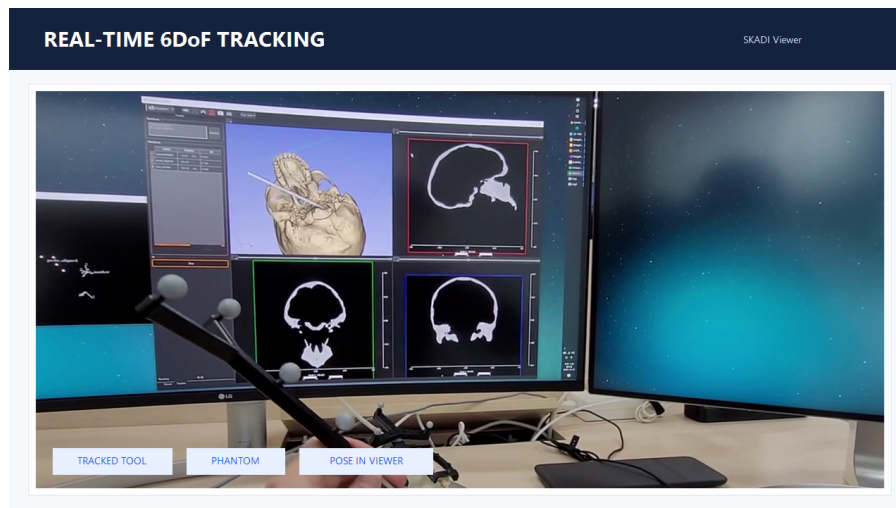


Figure 4. Real-time 6DoF marker tracking in SKADI Viewer

The lead demonstration and six public derivatives cover MarkerEditor, the API error model, Viewer 6DoF, de-identified dental registration, the Slicer template, and the robot-docking structure.

Define → Open → Track → Apply**Define**

Import, save, and export marker geometry in MarkerEditor, with validation on coordinate input.

Open

The application loads DtSkadi.dll and branches on the OpenEx() success or error result before entering an invalid state.

Track

Inspect real-time 6DoF marker pose and device state through the Viewer and API.

Apply

Surgical-navigation or robot precision-docking applications consume the validated transforms.

My role and API stability

Integrated DtSkadi.dll into applications, separated OpenEx() error branches so an open failure no longer continued into an invalid state and crash, and implemented MarkerEditor input validation. I maintain and release the SKADI Viewer family and API, public documentation, and integration support.

OpenEx() stability

In the DtSkadi.dll integration, OpenEx() success and error results are separated so failure stops before an invalid state and crash.

Viewer family and naming history

I maintain and release the SKADI Viewer family. The marker-editing tool MarkerEditor was renamed SkadiApp as its scope expanded.

[Public API change note](#)

Medical primary: surgical-navigation integration

In a de-identified dental CT and tray workflow, tracking coordinates are inspected through the Viewer and API and the application boundary is checked with the Slicer template.

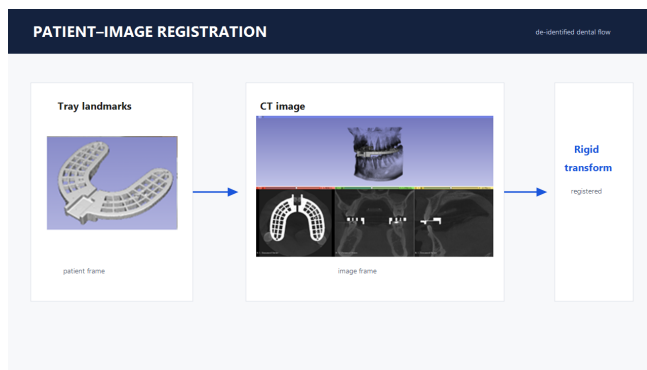


Figure 5. De-identified dental patient-to-image registration flow

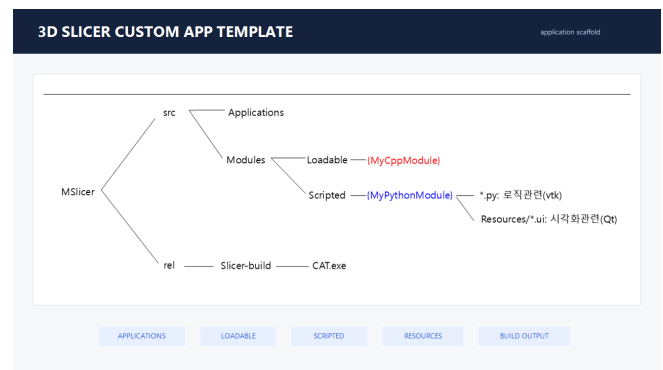


Figure 6. 3D Slicer custom-application template structure

My role

I own API and Viewer integration, public documentation and template maintenance, and integration support.

Team and partner boundary

The clinical workflow and final surgical-navigation application are team and partner results; no clinical effect is claimed as my result.

Industrial extension: robot precision docking

The same 6DoF tracking layer supplies the transform between robot-end and target markers for a docking application.

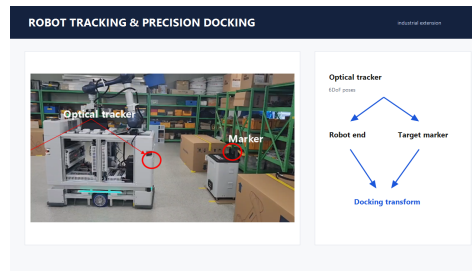


Figure 7. Coordinate structure for robot tracking and precision docking

My role

I own tracking-API and coordinate-integration support.

Team and partner boundary

Robot control and the final docking behavior are team and customer application results; product performance or customer outcomes are not claimed as mine.

Public documentation and product information

DOCUMENTATION

SKADI Viewer documentation

Explains device connection, marker editing, real-time tracking, and the Viewer workflow.

PRODUCT

SKADI Viewer product page

Public overview and application scope for the Viewer product family.

DOCUMENTATION

SKADI API documentation

Explains the public API entry point and integration flow used by tracking applications.

PRODUCT

SKADI API product page

Public product overview and application boundary for the SKADI API.

Limits and team result

The public material covers only software interfaces and integration boundaries. Faces, hospital and customer logos, licence paths, device IDs, internal figures, and performance assessments are excluded.

Device hardware, optical and mechanical design, final medical and industrial applications, robot control, sales, and customer outcomes belong to the team and partners. Product performance, clinical effects, and sales outcomes are not claimed as my results.

Integration boundary

I own the platform and integration support; final medical and industrial applications and product or customer outcomes belong to the team and partners.